

**Subject Name: Plant Tissue and Cell Culture**

**Subject No: BT20202**

**Prerequisites: Microbiology (BT20205) and Biochemistry (BT20203)**

### **Objective**

This course has been designed to give complete exposure to students on fundamental principles and application of plant growth *in vitro* (cell and tissue culture practices) leading to micropropagation and their scale up for industrial uses. Sequential information will be delivered from asepsis to nutrient formulation to culture initiation, as well as aspects of callus culture, regeneration and acclimatization. Adequate coverage on genetic modifications of plants leading to development of new transgenic plant lines has been ensured. Back up information on laboratory infrastructure, their uses and maintenance are provided. There will be matching laboratory classes on all aspects. The content will enable scientific excitement, sense of innovation and urge for applications.

**L-T-P 3-0-0**

**Credit 3**

### **Content**

1. Asepsis & culture initiation- theory, instruments & practices (3 lectures)
2. Nutrition & media- 4 nutritional categories, mineral nutrition, formulations in MS, B5 and N6 media (3 lectures)
3. Fundamentals of plant cell & tissue culture- culture initiation, callus culture, organogenesis, plant regeneration (3 lectures)
4. Clonal propagation and its industrial applications- enhanced axillary branching, adventitious budding & somatic embryogenesis (4 lectures)
5. Synthetic seed technology (2 lectures)
6. Use of liquid media, automation and artificial intelligence (3 lectures)
7. Somaclonal variations: Definition, scheme and applications of obtaining somaclonal variations; Various factors/parameters influencing somaclonal variations; Molecular mechanisms behind somaclonal variations- several genetic and epigenetic changes (3 lectures)
8. Protoplasts isolation and culture: Importance of protoplast culture; Principle/procedure of protoplast isolation; Different techniques of protoplast culture and their advantages and disadvantages (3 lectures)
9. Somatic hybridization: Different methods of protoplast fusion; Somatic hybrid, cytoplasmic hybrid or cybird; Identification and screening of hybrid cells/fusion products by different methods/strategies; Verification and characterization of hybrid plants (3 lectures)
10. Gene transfer to plant system or development of genetically modified (GM) or genetically engineered (GE) plants: Molecular basis of GM/GE plant development; Various methods of gene delivery/ transformation techniques; Verification and characterization of transformed or transgenic plant lines (3 lectures)
11. Green house and field protocols for tissue cultured and GM crops (2 lectures)

## 12. Indian Institutions and industries making monumental contributions (3 lectures)

### **Text books -**

1. Introduction to Plant Tissue Culture by M.K. Razdan Publisher- Oxford & IBH, 3rd edn ISBN: 8120417933
2. Introduction to Plant Biotechnology by H. S. Chawla Oxford and IBH, 3rd edn ISBN: 9781138407671
3. Plant Tissue Culture: An Introductory Text By Sant Saran Bhojwani& Prem Kumar Dantu Publisher- Springer Science & Business Media ISBN 978-81-322-1025-2
4. Plant Biotechnology by Slatter, Scott & Fowler Publisher- Oxford ISBN 9780199282616, 0199282617

### **Further reading references-**

1. Capell et al. Potential Applications of Plant Biotechnology against SARS-CoV-2. Trends in Plant Science, 25 (7), 635-643 (2020)
2. Lu et al. Precise genome modification in tomato using an improved prime editing system. Plant Biotechnology Journal 19, 415–417 (2021)
3. Zilberman &Lefler. Biotechnology for African food security. Nature Food 2, 79 (2021)